scientific reports

Check for updates

OPEN

Cardiovascular disease prediction using variational recurrent autoencoders with uncertainty estimation

Ashim Chandra Das[1], Md Shujan Shak[1], Nabila Rahman[2], Fuad Mahmud[3], Hashibul Ahsan Shoaib[4] & Md. Jakir Hossen[5]✉

Cardiovascular diseases (CVDs) remain a leading cause of mortality worldwide, motivating reproducible benchmark studies on accurate classification methods with calibrated probability estimates. In this paper, we propose a deep learning framework based on a Variational Recurrent Autoencoder (VRAE) with uncertainty estimation for CVD classification from static tabular clinical records represented as synthetic noise-augmented pseudo-sequences. The two datasets used in this study are static tabular datasets rather than real longitudinal clinical time-series. Therefore, the constructed pseudo-sequence dimension is used for denoising latent representation learning under simulated measurement perturbation and missingness, not for modeling observed clinical temporal dependencies, patient trajectories, disease progression, or treatment dynamics. The model integrates Gated Recurrent Unit (GRU)-based encoding, variational latent sampling, and Monte Carlo (MC) dropout to learn robust latent representations and quantify predictive uncertainty. We evaluate the model on two publicly available datasets: the Heart Failure Prediction dataset (918 samples) and the Cardiovascular Disease dataset (70,000 samples). The model achieves an accuracy of 95.8% and F1-score of 95.7% on the heart failure dataset, and an accuracy of 96.1% with F1-score of 96.0% on the larger cardiovascular dataset, outperforming traditional classifiers, calibrated tabular baselines, and deep learning baselines including regularized logistic regression, calibrated XGBoost, calibrated LightGBM, calibrated CatBoost, Long Short-Term Memory (LSTM), and GRU-Attention. The proposed VRAE model also demonstrates robustness under synthetic noise and provides improved calibration, as indicated by the lowest Brier scores (0.061 and 0.059) across both datasets. Additional calibration, selective prediction, and threshold-based utility analyses support the reliability of the benchmark results. These findings support uncertainty-aware representation learning for benchmark-level static tabular CVD classification, while independent validation on real-world longitudinal electronic health record cohorts is required before claims about prospective risk prediction, early intervention, clinical time-series prediction, or clinical deployment can be made.

Keywords Cardiovascular disease prediction, Variational recurrent autoencoder, Uncertainty estimation, Deep learning, Monte Carlo dropout, SHAP, Healthcare AI

Cardiovascular diseases (CVDs) are the leading cause of death globally, with an estimated 19.8 million deaths in 2022, representing approximately 32% of all global deaths according to the World Health Organization[1]. Accurate CVD classification and calibrated risk-factor assessment from structured clinical records can support reproducible benchmark evaluation and model development. Traditional risk scoring systems, such as the Framingham Risk Score and the ASCVD Risk Estimator, estimate cardiovascular risk using predefined clinical variables and population-derived equations[2,3]. Although these tools are clinically useful and interpretable, they rely on static risk-factor representations that may not fully capture nonlinear interactions, patient heterogeneity, or uncertainty in individual predictions. In recent years, machine learning (ML) and deep learning (DL)

[1]University of the Potomac, Washington, VA 22043, USA. [2]Trine University, Detroit 48101, USA. [3]Information Assurance and Cybersecurity, Gannon University, Erie 16541, USA. [4]Information Technology, St. Francis College, Brooklyn 11201, USA. [5]Center for Advanced Analytics (CAA), Faculty of Engineering & Technology (FET), Multimedia University, 75450 Melaka, Malaysia. ✉email: jakir.hossen@mmu.edu.my

techniques have gained traction in clinical prediction research due to their ability to uncover nonlinear patterns and complex associations in high-dimensional data[4,5].

Despite notable progress, many ML and DL models for cardiovascular risk prediction suffer from critical limitations[6,7]. Many approaches assume static, clean datasets and do not model uncertainty in predictions, which limits reliability assessment in healthcare prediction research. Clinical data, especially from electronic health records (EHRs), are often noisy, incomplete, and irregularly sampled[8,9]. Moreover, many existing models operate as black boxes and provide point predictions without expressing confidence levels, limiting the reliability of their outputs in benchmark-level healthcare prediction studies. These challenges necessitate models that provide accurate predictions, account for uncertainty, and remain stable under noisy or incomplete structured clinical inputs.

Recurrent neural networks (RNNs), particularly Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models, have shown promise when ordered clinical observations are available[10,11]. However, the datasets used in this study do not contain observed longitudinal visits or clinical time-series. Variational autoencoders (VAEs)[12] provide a probabilistic framework for learning latent representations, enabling them to handle noisy inputs and generate calibrated predictions. When combined with recurrent encoders, they form Variational Recurrent Autoencoders (VRAEs), which can encode sequence-structured inputs while modeling latent distributions. In this study, the sequence structure is synthetically generated from static tabular records through controlled perturbation and masking. Thus, the recurrent encoder is used for noise-augmented pseudo-sequence representation learning rather than for learning real temporal dependencies, disease progression, treatment effects, or patient trajectories.

Several key observations motivated this work. First, structured clinical datasets are often noisy, incomplete, and heterogeneous, even when available as static tabular records. This motivates a representation learning strategy that remains stable under simulated measurement variation and missingness. Second, traditional classifiers and many deep learning models provide point predictions without predictive confidence, which limits reliability assessment in risk-sensitive healthcare prediction research. Third, the sequence dimension used in this study is synthetically constructed from static records and is used for noise-augmented representation learning rather than for modeling observed patient visits, treatment effects, disease progression, or true longitudinal trajectories. Lastly, model interpretation is important for assessing whether predictions are driven by clinically plausible variables. By using SHAP, the model predictions can be inspected in relation to clinically meaningful features without claiming validated clinical decision-support readiness.

In this paper, we address these limitations by proposing a VRAE-based architecture for CVD classification from static tabular clinical datasets represented as synthetic noise-augmented pseudo-sequences. The constructed pseudo-sequence representation is used to support denoising and uncertainty-aware latent representation learning, not to claim observed longitudinal clinical time-series modeling, temporal-dependency learning, or patient trajectory modeling. By incorporating Monte Carlo (MC) dropout at the inference stage, the model quantifies predictive uncertainty. Furthermore, SHAP (SHapley Additive exPlanations) values are used to identify key contributing clinical features and improve the transparency of model predictions. The main contributions of this paper are as follows:

- We propose a deep learning framework based on a Variational Recurrent Autoencoder (VRAE) with Monte Carlo (MC) dropout for uncertainty-aware CVD classification from static tabular clinical records.
- We introduce a synthetic noise-augmented pseudo-sequence construction strategy that expands each static record into repeated perturbed views, enabling denoising representation learning while avoiding claims of true longitudinal trajectory or temporal-dependency modeling.
- We compare the proposed model with classical and deep learning baselines across multiple performance metrics, including accuracy, precision, recall, F1-score, AUC, calibration, and statistical significance.
- We evaluate robustness under increasing levels of Gaussian noise and missing-value masking, showing the model's capacity to handle simulated measurement variation and incomplete inputs.
- We employ SHAP-based feature attribution to interpret model decisions and identify clinically relevant factors, enhancing model transparency and alignment with medical knowledge.
- We assess computational efficiency to examine the trade-off between predictive performance, uncertainty estimation, and resource usage in benchmark-level static tabular classification.

The remainder of this paper is organized as follows. The Related Work section reviews relevant literature on cardiovascular disease prediction, deep learning in healthcare, and uncertainty-aware models. The Methodology section describes the data preprocessing pipeline and the proposed VRAE-based framework, including the architectural components and training strategy. The Results section presents comprehensive experimental evaluations, including performance comparisons, calibration analysis, uncertainty analysis, robustness testing, and explainability. The Discussion section examines the implications and limitations of the findings. Finally, the Conclusions section summarizes the main findings and outlines the need for external validation.

Related work

Predictive modeling for cardiovascular diseases has received significant attention in recent years, driven by the availability of clinical data and the evolution of machine learning techniques. This section outlines existing research from several complementary perspectives: traditional statistical methods, classical machine learning models, deep learning architectures, time-series approaches, and uncertainty-aware prediction in healthcare. The goal is to position our proposed approach within the context of this evolving landscape.

Historically, cardiovascular risk assessment has relied on statistical tools such as the Framingham Risk Score, the ASCVD Risk Calculator, and the SCORE model[2,13,14]. These methods typically use a limited set of

features such as age, blood pressure, cholesterol, smoking status, and diabetes history to compute a risk score. While clinically interpretable, these approaches are constrained by predefined variable sets and may not fully model complex, nonlinear interactions between variables. Related survival-analysis studies have also developed prognostic risk score (PRS) systems for recurrent or treatment-oriented diseases. For example, regularized Cox proportional hazards (RCPH) models have been used to identify prognostic biomarkers and construct PRS systems through univariate and multivariate Cox regression analysis[15]. More recently, CNet-Cox extended Cox-based survival modeling by incorporating network regularization for interpretable biomarker discovery and PRS-based risk stratification[16]. These PRS methods are valuable for time-to-event prognosis, whereas the present study focuses on binary cardiovascular disease prediction with uncertainty-aware deep representation learning.

As access to structured health records improved, a wide range of supervised learning techniques were applied to cardiovascular prediction[17]. Logistic regression and support vector machines (SVM) have been frequently used due to their simplicity and effectiveness in high-dimensional settings[18]. Decision trees, random forests, and gradient boosting machines like XGBoost and LightGBM have also been applied extensively, offering better accuracy and handling of missing data[19]. Several studies employed k-nearest neighbors (KNN), naïve Bayes, and ensemble methods for feature-rich datasets[20]. Feature selection techniques such as recursive feature elimination (RFE), mutual information, and principal component analysis (PCA) have also been used to improve model efficiency[21]. Recent biomedical machine learning research has further emphasized stable feature selection. For example, StabML-RFE combines eight machine learning-based RFE variants, including AdaBoost, decision tree, gradient-boosted decision tree, naïve Bayes, neural network, random forest, support vector machine, and XGBoost, within an aggregation framework to improve the stability of selected biomedical features[22]. Although these methods improve feature ranking and model efficiency, they generally assume independent and identically distributed samples and do not directly model temporal structure or predictive uncertainty.

Deep learning models have recently gained prominence in clinical prediction tasks due to their capacity to model nonlinear interactions and automatically learn hierarchical representations from raw data. Feedforward deep neural networks (DNNs) have been employed to predict heart disease based on EHR features[23]. Convolutional neural networks (CNNs), particularly 1D CNNs, have been used for structured data, while 2D CNNs are often applied to imaging modalities such as echocardiograms and CT angiograms[24,25]. Recurrent neural networks (RNNs), including Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models, have been particularly effective for modeling sequential health data such as patient visits, medication records, and symptom trajectories[26]. Bidirectional variants of LSTM and GRU further enhance temporal context by incorporating both past and future dependencies. Attention mechanisms and Transformer-based architectures have recently been introduced into medical applications, providing dynamic feature weighting and improved sequence representation[27]. These include models like RETAIN (Reverse Time Attention), which uses attention to highlight the most relevant clinical events for prediction, and Med-BERT, which adapts BERT-style architectures to EHR data. However, these models often require large training datasets and can be sensitive to noise and missingness, common in real-world clinical environments.

Temporal modeling is important in healthcare when real longitudinal observations are available, such as patient visits, medication records, laboratory trends, or disease-monitoring trajectories. In such settings, LSTM, GRU, temporal convolutional networks (TCNs), sequence-to-sequence models, and hierarchical RNNs have been applied to tasks such as ICU mortality prediction, heart failure onset detection, and sepsis forecasting[28]. Multivariate time-series modeling has also been explored using dynamic Bayesian networks, hidden Markov models (HMMs), and autoregressive integrated moving average (ARIMA) models[29]. These approaches are designed for observed temporal data and require time-stamped events or repeated measurements.

The present study does not use such longitudinal data. Instead, it uses static tabular records and constructs repeated perturbed views of each record for denoising representation learning. Therefore, the proposed pseudo-sequence construction should not be grouped with clinical trajectory generation methods or interpreted as modeling true temporal progression.

Despite high accuracy, deep models are often criticized for their lack of calibration and inability to express uncertainty. This is particularly critical in healthcare, where overconfident wrong predictions can lead to harmful decisions. To address this, several techniques for uncertainty modeling have been proposed. Bayesian neural networks place distributions over weights and use variational inference to capture model uncertainty. Monte Carlo dropout, introduced as an approximation to Bayesian inference, is widely used due to its simplicity and computational feasibility. Deep ensembles, where multiple independently trained models provide a distribution over predictions, have also been explored. Temperature scaling, Platt scaling, and isotonic regression are post-hoc calibration methods used to improve probability estimates from classifiers[30]. These techniques, while useful, often treat uncertainty estimation as an afterthought rather than integrating it into the modeling process. Our approach incorporates uncertainty directly into the architecture via variational sampling and MC dropout, improving robustness and interpretability.

In summary, while significant advances have been made in the application of machine learning and deep learning to cardiovascular disease prediction, critical challenges remain. Many models for static tabular clinical data do not jointly address robustness to input perturbation, uncertainty estimation, and post-hoc explanation. True longitudinal risk prediction remains a separate setting that requires time-stamped visits, a defined index time, a look-back window, a prediction horizon, and future outcome labels. The proposed method addresses the static tabular setting by combining synthetic noise-augmented pseudo-sequence representation, variational latent learning, MC dropout-based uncertainty estimation, and SHAP-based post-hoc explanation. Therefore, the contribution is positioned as uncertainty-aware denoising representation learning for static CVD classification rather than as a model of real clinical temporal dynamics.

Methodology

This section describes the proposed methodology for cardiovascular disease classification using a Variational Recurrent Autoencoder (VRAE) with uncertainty estimation. The input data consist of static tabular clinical records that are converted into synthetic noise-augmented pseudo-sequences for denoising representation learning. These constructed pseudo-sequences do not represent observed longitudinal visits, treatment stages, or disease progression. The architecture consists of a recurrent encoder, a latent representation with variational inference, uncertainty modeling using Monte Carlo (MC) dropout, and a classification head, as shown in Fig. 1. The model is trained in an end-to-end fashion using both reconstruction and classification objectives.

Data preprocessing

To prepare the data for training the Variational Recurrent Autoencoder (VRAE), we performed preprocessing steps that converted static clinical records into synthetic noise-augmented sequence representations and handled missing or noisy values. Two datasets were used: (i) the Heart Failure Prediction Dataset (918 samples, 11 clinical features) and (ii) the Cardiovascular Disease Dataset (70,000 samples). Both datasets are static tabular datasets rather than observed longitudinal electronic health record cohorts.

Feature selection and encoding

We selected clinically relevant features common to both datasets. Categorical features such as `Sex`, `ChestPainType`, `RestingECG`, `ExerciseAngina`, and `ST_Slope` were one-hot encoded. Numerical features were normalized using min-max scaling:

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{1}$$

where x is the original feature value, and $x_{\min}$, $x_{\max}$ are the minimum and maximum values of the feature.

Synthetic noise-augmented pseudo-sequence construction

Because both datasets contain static tabular records rather than longitudinal patient visits, the sequence dimension was synthetically constructed for noise-augmented representation learning. Each original record was expanded into a pseudo-sequence $\mathrm{x}_{1:T} \in \mathbb{R}^{T \times F}$, where $T = 5$ denotes the number of perturbed views and F denotes the number of features. The index t represents a synthetic perturbation step rather than a real clinical visit, follow-up interval, laboratory interval, treatment phase, or disease progression point.

The purpose of this construction is not to create a clinically meaningful temporal axis. Instead, it creates multiple stochastic views of the same static record so that the encoder-decoder model can learn a latent representation that is stable under plausible measurement perturbation and feature-level missingness. This design is appropriate for evaluating denoising representation learning and robustness under controlled input corruption, because the model is trained to reconstruct the perturbed pseudo-sequence while preserving the cross-sectional CVD label. The selected length $T = 5$ provides repeated perturbed views for reconstruction without excessively duplicating each static sample or substantially increasing computational cost.

For each normalized continuous feature x_j, Gaussian perturbation was applied to create controlled variation across the synthetic pseudo-sequence:

$$x_{t,j} = \text{clip}(x_j + \epsilon_{t,j}, 0, 1), \quad \epsilon_{t,j} \sim \mathcal{N}(0, \sigma_j^2), \quad \sigma_j = \alpha s_j \tag{2}$$

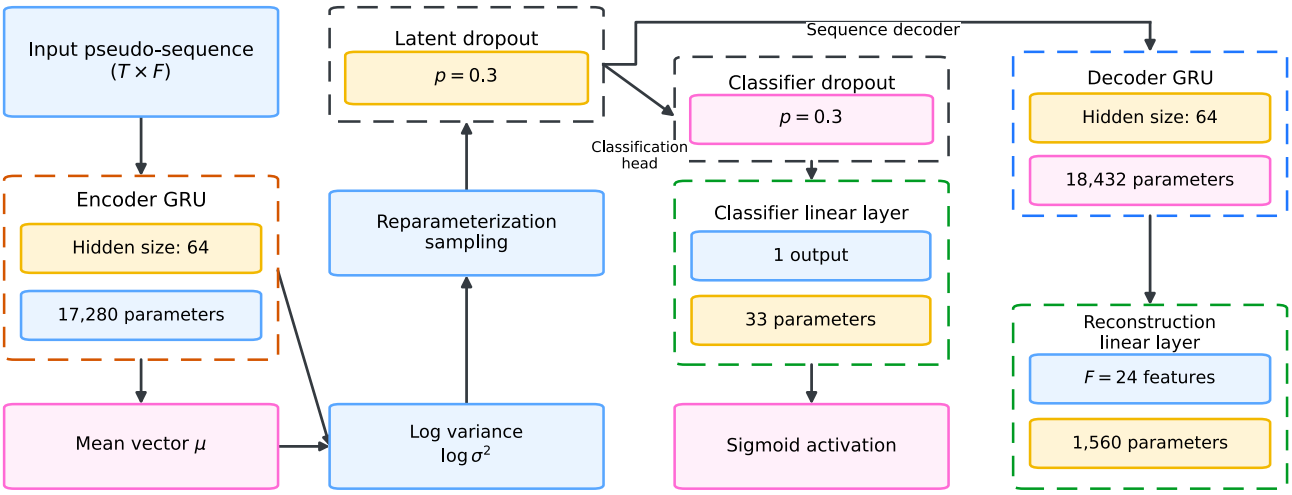


Fig. 1. Overview of the proposed VRAE framework for CVD classification from static tabular records represented as synthetic noise-augmented pseudo-sequences.

where s_j is the standard deviation of feature j estimated from the training split and $\alpha = 0.05$ is the pseudo-sequence perturbation noise scale used in the main experiments. The $\mathrm{clip}(\cdot)$ operation keeps perturbed normalized values within the valid range $[0, 1]$. One-hot encoded categorical variables were not jittered to avoid generating invalid category values. Random masking was then applied to selected feature values to simulate incomplete tabular inputs. Therefore, the synthetic pseudo-sequence construction should be interpreted as a controlled augmentation and denoising strategy for static clinical records. It does not establish real patient trajectories, clinically meaningful temporal dependencies, disease progression dynamics, treatment effects, or prospective risk evolution.

Missing data simulation
To test robustness, we introduced missing values into the synthetic pseudo-sequence data with a masking probability $p_{\mathrm{miss}} = 0.10$ in the main experiments. A binary mask $m_{t,j}^{(i)} \in \{0, 1\}$ was sampled independently per perturbed view and feature:

$$\tilde{x}_{t,j}^{(i)} = m_{t,j}^{(i)} x_{t,j}^{(i)} + \left(1 - m_{t,j}^{(i)}\right) \cdot \bar{x}_{j,\mathrm{train}} \tag{3}$$

This simulates feature-level missingness in static clinical records. Masking was applied independently within each split after the patient-record-level split. Missing entries were replaced using feature means estimated from the training split, and the model was trained to reconstruct the complete augmented input despite masked entries.

Final data structure
After preprocessing, each patient record was represented as a synthetic multivariate pseudo-sequence of perturbed views:

$$\mathrm{X} = [\mathrm{x}_1, \mathrm{x}_2, \ldots, \mathrm{x}_T], \quad \mathrm{x}_t \in \mathbb{R}^F \tag{4}$$

The target variable for each sequence was the presence or absence of cardiovascular disease, $y \in \{0, 1\}$, replicated across the sequence for supervised training.

Clinical task definition
The prediction task is defined as binary cross-sectional CVD classification from a static clinical record. The label indicates the presence or absence of CVD in the corresponding benchmark dataset. Because the datasets do not contain observed longitudinal visits, index times, look-back windows, treatment timelines, prediction horizons, or future outcome windows, the task should not be interpreted as prospective risk prediction or early detection over time. The replicated label is used only to supervise the repeated perturbed views of the same static record.

Data splitting and batching
Each dataset was first split into training (70%), validation (15%), and test (15%) sets at the original patient-record level before normalization, imputation, masking, or synthetic pseudo-sequence construction. This split-before-augmentation design was used to prevent data leakage across partitions. All $T = 5$ perturbed views generated from the same original patient record were kept within the same assigned split, and no synthetic or masked version of an original record was allowed to appear in a different split. Min-max scaling parameters, feature means used for masked-value replacement, and model-selection thresholds were estimated only from the training split and then applied unchanged to the validation and test splits. The data were then batched as fixed-size synthetic pseudo-sequences, and each batch $\mathcal{B}$ consisted of N original records:

$$\mathcal{B} = \left\{ (\mathrm{X}^{(i)}, y^{(i)}) \right\}_{i=1}^{N} \tag{5}$$

Because each original record was expanded into a fixed number of perturbed views, $T = 5$, no cross-split sliding windows or cross-record sequence construction were used.

Proposed methodology

Figure 2 presents an overview of the proposed Variational Recurrent Autoencoder (VRAE) methodology for predicting cardiovascular disease. This section describes the methodology in greater detail.

Variational recurrent autoencoder (VRAE)
The VRAE is designed to process synthetic noise-augmented pseudo-sequences derived from static clinical records and to learn a compact latent representation that is robust to perturbation and missingness. Given an input pseudo-sequence $\mathrm{X} = [\mathrm{x}_1, \mathrm{x}_2, \ldots, \mathrm{x}_T]$, where each $\mathrm{x}_t \in \mathbb{R}^F$, the VRAE encodes the repeated perturbed views into a latent distribution $q(\mathrm{z}|\mathrm{X})$. In this study, the recurrent encoder summarizes stochastic views of the same static record rather than observed longitudinal clinical events, so the learned representation should be interpreted as denoising latent representation learning rather than temporal-dynamics modeling.

Encoder
The encoder is a recurrent neural network (RNN), specifically a gated recurrent unit (GRU), that outputs the hidden state h_T after reading the entire sequence:

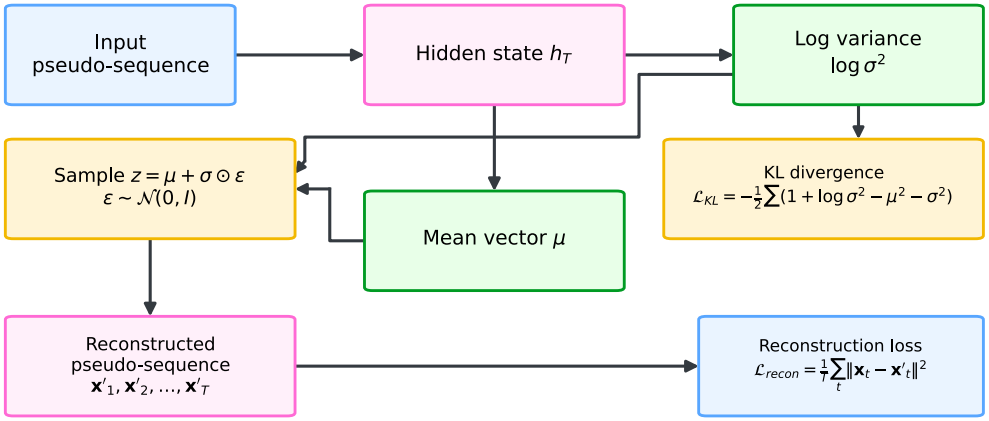


Fig. 2. Latent representation and reconstruction process of the proposed VRAE for synthetic noise-augmented pseudo-sequences.

$$\mathrm{h}_t = \mathrm{GRU}(\mathrm{x}_t, \mathrm{h}_{t-1}) \tag{6}$$

$$\mathrm{h}_T = \mathrm{EncoderRNN}(\mathrm{X}) \tag{7}$$

From the final hidden state h_T, we derive the mean $\boldsymbol{\mu}$ and standard deviation $\boldsymbol{\sigma}$ of the approximate posterior:

$$\boldsymbol{\mu} = \mathrm{W}_\mu \mathrm{h}_T + \mathrm{b}_\mu \tag{8}$$

$$\log(\boldsymbol{\sigma}) = \mathrm{W}_\sigma \mathrm{h}_T + \mathrm{b}_\sigma \tag{9}$$

A latent vector $\mathrm{z} \in \mathbb{R}^d$ is sampled using the reparameterization trick:

$$\mathrm{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathrm{I}) \tag{10}$$

Decoder
The decoder is another GRU that reconstructs the original sequence from the latent code:

$$\hat{\mathrm{x}}_t = \mathrm{DecoderGRU}(\mathrm{z}, \hat{\mathrm{x}}_{t-1}) \tag{11}$$

The output is a sequence $\hat{\mathrm{X}} = [\hat{\mathrm{x}}_1, \hat{\mathrm{x}}_2, \ldots, \hat{\mathrm{x}}_T]$, which is compared to the original sequence using a mean squared error (MSE) loss:

$$\mathcal{L}_{\mathrm{recon}} = \frac{1}{T} \sum_{t=1}^{T} \|\mathrm{x}_t - \hat{\mathrm{x}}_t\|_2^2 \tag{12}$$

KL divergence loss
To regularize the latent space, we use the Kullback-Leibler (KL) divergence between the approximate posterior $q(\mathrm{z}|\mathrm{X})$ and the prior $p(\mathrm{z}) = \mathcal{N}(0, \mathrm{I})$:

$$\mathcal{L}_{\mathrm{KL}} = -\frac{1}{2} \sum_{i=1}^{d} \left(1 + \log(\sigma_i^2) - \mu_i^2 - \sigma_i^2\right) \tag{13}$$

Uncertainty estimation with MC dropout
To estimate predictive uncertainty under noisy inputs, we apply Monte Carlo dropout at inference time. Specifically, dropout is applied to the latent representation z and to the classification head. During inference, $M = 30$ stochastic forward passes are made in the main experiments:

$$\{\hat{y}^{(1)}, \hat{y}^{(2)}, \ldots, \hat{y}^{(M)}\} = \mathrm{Classifier}(\mathrm{Dropout}(\mathrm{z})) \tag{14}$$

The final prediction is obtained by averaging:

$$\bar{y} = \frac{1}{M} \sum_{i=1}^{M} \hat{y}^{(i)} \tag{15}$$

Predictive uncertainty is estimated via variance:

$$\mathrm{Var}(\hat{y}) = \frac{1}{M}\sum_{i=1}^{M}(\hat{y}^{(i)} - \bar{y})^2 \tag{16}$$

For reliability-oriented interpretation, the model reports both the mean predicted probability $\bar{y}$ and the predictive uncertainty $u = \mathrm{Var}(\hat{y})$. A validation-calibrated decision rule can be defined as follows:

$$\text{Decision} = \begin{cases} \text{Positive classification with low uncertainty}, & \bar{y} \geq \tau_p \text{ and } u \leq \tau_u, \\ \text{Negative classification with low uncertainty}, & \bar{y} < \tau_p \text{ and } u \leq \tau_u, \\ \text{Uncertain classification for review}, & u > \tau_u. \end{cases} \tag{17}$$

here $\tau_p = 0.5$ is used as the default binary classification threshold, while τ_u is selected from the validation-set uncertainty distribution. In this study, the 75th percentile of validation uncertainty is used for selective prediction analysis, and the 90th percentile is examined as a stricter review trigger. High uncertainty should not be interpreted as high disease risk; it indicates that the prediction is unstable across stochastic forward passes and should be reviewed with input data quality and clinical context.

Classification head
The latent vector z is passed through a fully connected classifier to predict the presence of cardiovascular disease:

$$\hat{y} = \sigma(\mathrm{W}_c\mathrm{z} + b_c) \tag{18}$$

where σ is the sigmoid activation function.

The binary cross-entropy (BCE) loss is used for training the classifier:

$$\mathcal{L}_{\text{class}} = -y\log(\hat{y}) - (1-y)\log(1-\hat{y}) \tag{19}$$

Overall objective function
The total loss combines reconstruction, KL divergence, and classification loss:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{recon}}\mathcal{L}_{\text{recon}} + \lambda_{\text{KL}}\mathcal{L}_{\text{KL}} + \lambda_{\text{class}}\mathcal{L}_{\text{class}} \tag{20}$$

Hyperparameters $\lambda_{\text{recon}}, \lambda_{\text{KL}}, \lambda_{\text{class}}$ are tuned via validation.

In the main experiments, the loss weights were set to $\lambda_{\text{recon}} = 1.0$, $\lambda_{\text{KL}} = 0.001$, and $\lambda_{\text{class}} = 1.0$. The KL term was assigned a smaller weight to regularize the latent space without dominating the supervised classification objective. These values were selected using validation-set performance.

Architectural details

The proposed model architecture is composed of three main components: the encoder, the latent space with variational sampling, and the decoder followed by a classification head. The architecture was implemented using PyTorch, and hyperparameters were selected based on cross-validation performance. Dropout layers were incorporated in the latent space and classifier for Monte Carlo (MC) dropout-based uncertainty estimation. Table 1 provides a detailed breakdown of the architecture:

Encoder
The encoder is a single-layer GRU network with a hidden size of 64. It encodes the entire sequence into a fixed-dimensional hidden state vector h_T. This state is passed through two linear layers to generate the mean $\boldsymbol{\mu}$ and log-variance $\log(\boldsymbol{\sigma})$ of the latent space.

Layer	Input Dimension	Output Dimension	Parameters
Input Sequence	$T \times F$	–	–
Encoder GRU	24	64	17, 280
Linear to $\boldsymbol{\mu}$	64	32	2080
Linear to $\log \boldsymbol{\sigma}$	64	32	2080
Sampling	32	32	–
Dropout (latent)	32	32	–
Decoder GRU	32	64	18, 432
Linear reconstruction	64	24	1, 560
Dropout (classifier)	32	32	–
Linear classification	32	1	33
Sigmoid Activation	1	1	–
Total Parameters	–	–	**41,465**

Table 1. Summary of Model Architecture.

Latent space and sampling
The latent space dimension d is set to 32. Sampling is performed using the reparameterization trick to allow backpropagation. Dropout with probability 0.3 is applied to the latent vector during training and inference to enable uncertainty estimation via Monte Carlo sampling.

Decoder
The decoder is also a single-layer GRU that takes the sampled latent vector as its initial hidden state and reconstructs the sequence. The output of the decoder is passed through a linear layer to map the hidden state back to the original feature dimension.

Classification head
A separate classification head takes the latent vector z, applies dropout, and then passes it through a fully connected layer followed by a sigmoid activation to predict the probability of cardiovascular disease.

The model balances complexity with computational efficiency, making it suitable for learning robust representations from static tabular clinical records under synthetic noise and missing-value perturbation.

Algorithm summary

Algorithm 1 summarizes the proposed pipeline for cardiovascular disease prediction using a Variational Recurrent Autoencoder with uncertainty estimation. The algorithm begins with preprocessing the input clinical dataset, where numerical features are normalized, categorical variables are one-hot encoded, and each static patient record is expanded into a synthetic noise-augmented sequence of fixed length T using Gaussian perturbation and random masking. The generated pseudo-sequence represents repeated perturbed views of the same static record rather than observed clinical visits, laboratory intervals, treatment phases, or disease progression points. The resulting sequences are passed through a gated recurrent unit (GRU) encoder to produce a hidden representation, from which the mean $\boldsymbol{\mu}$ and log-variance $\log(\boldsymbol{\sigma})$ of the latent distribution are derived. A latent vector z is sampled using the reparameterization trick. The decoder, also a GRU, reconstructs the augmented input sequence from z, and the reconstruction loss is computed using mean squared error. Simultaneously, the model computes the Kullback-Leibler divergence to regularize the latent space. For classification, the latent vector z is passed through a Monte Carlo dropout layer multiple times to generate M stochastic predictions. The mean of these predictions provides the final probability of cardiovascular disease, while their variance estimates the model's uncertainty. The total loss is a weighted sum of reconstruction, KL divergence, and classification losses, and the network is trained end-to-end via backpropagation.

Input: Static clinical dataset $D = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^{N}$, number of perturbed views T, latent dimension d
Output: Predicted CVD probability $\bar{y}$ and predictive uncertainty $\mathrm{Var}(\hat{y})$
Preprocessing: for *each sample* $\mathbf{x}^{(i)} \in D$ **do**
    Normalize numerical features
One-hot encode categorical features
Construct synthetic noise-augmented pseudo-sequence $\mathbf{X}^{(i)} = [\mathbf{x}_1, \ldots, \mathbf{x}_T]$ using Gaussian perturbation
Apply random masking to simulate missing values
Encoding:
Pass $\mathbf{X}^{(i)}$ through GRU encoder to get $\mathbf{h}_T$
Compute $\mu, \log(\sigma)$ from $\mathbf{h}_T$
Sample latent code $\mathbf{z} = \mu + \sigma \odot \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$
Decoding and Reconstruction:
Decode $\mathbf{z}$ into reconstructed sequence $\hat{\mathbf{X}}$
Compute reconstruction loss $\mathscr{L}_{\text{recon}}$
Compute KL divergence $\mathscr{L}_{\text{KL}}$
Classification: for $i = 1$ *to M (MC Dropout samples)* **do**
    Apply dropout to $\mathbf{z}$ to get $\mathbf{z}^{(i)}$
Predict $\hat{y}^{(i)} = \mathrm{sigmoid}(\mathbf{W}_c \mathbf{z}^{(i)} + b_c)$
Compute mean prediction $\bar{y} = \frac{1}{M} \sum_{i=1}^{M} \hat{y}^{(i)}$
Estimate uncertainty $\mathrm{Var}(\hat{y}) = \frac{1}{M} \sum_{i=1}^{M} (\hat{y}^{(i)} - \bar{y})^2$
Training Objective:
Compute total loss:

$$\mathscr{L}_{\text{total}} = \lambda_{\text{recon}} \mathscr{L}_{\text{recon}} + \lambda_{\text{KL}} \mathscr{L}_{\text{KL}} + \lambda_{\text{class}} \mathscr{L}_{\text{class}}$$

Update model parameters using backpropagation

Algorithm 1. Variational Recurrent Autoencoder for Cardiovascular Disease Prediction

Training procedure

The model was trained for a maximum of 100 epochs using the Adam optimizer with a learning rate of 1×10^{-3}, batch size of 64, and dropout probability of 0.3. Early stopping was applied using validation loss with a patience of 15 epochs and a minimum improvement threshold of 1×10^{-4}; the checkpoint with the lowest validation

loss was restored for testing. During inference, Monte Carlo dropout used $M = 30$ stochastic forward passes to compute the mean prediction and predictive variance. Repeated-run experiments used five random seeds: 42, 43, 44, 45, and 46. Hyperparameter search was performed on the validation set using the following candidate ranges: hidden size $\{32, 64, 128\}$, latent dimension $\{16, 32, 64\}$, learning rate $\{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}\}$, batch size $\{32, 64, 128\}$, dropout probability $\{0.1, 0.2, 0.3, 0.4, 0.5\}$, KL loss weight $\{1 \times 10^{-4}, 1 \times 10^{-3}, 1 \times 10^{-2}\}$, reconstruction loss weight $\{0.5, 1.0\}$, classification loss weight $\{1.0\}$, and MC dropout samples $\{10, 30, 50\}$. The selected configuration was hidden size 64, latent dimension 32, learning rate 1×10^{-3}, batch size 64, dropout 0.3, $\lambda_{\text{recon}} = 1.0$, $\lambda_{\text{KL}} = 0.001$, $\lambda_{\text{class}} = 1.0$, and $M = 30$. All experiments were implemented in Python 3.10 using PyTorch, NumPy, scikit-learn, SHAP, XGBoost, LightGBM, and CatBoost, and the models were trained on a single GPU with 12 GB memory.

Results

This section presents the experimental results of the proposed VRAE model on the Heart Failure and Cardiovascular Disease datasets. The task is evaluated as static tabular CVD classification under synthetic noise augmentation. We compare the proposed model with classical machine learning models, calibrated tabular baselines, and deep learning baselines using accuracy, precision, recall, F1-score, AUC, confidence intervals, repeated-run variability, calibration diagnostics, selective prediction, and threshold-based clinical utility analysis.

Baseline evaluation protocol

All baseline models were evaluated using the same patient-record-level training, validation, and test splits used for the proposed VRAE. No test-set information was used during preprocessing, model selection, calibration, or threshold selection. Numerical normalization parameters and feature-imputation statistics were estimated only from the training split and then applied unchanged to the validation and test splits. Categorical variables were one-hot encoded using categories identified from the training data.

Classical and tabular machine learning baselines, including regularized logistic regression, random forest, XGBoost, calibrated XGBoost, calibrated LightGBM, and calibrated CatBoost, were trained on the static tabular feature representation after the same encoding and scaling pipeline. Sequence-based deep learning baselines, including 1D CNN, LSTM, BiLSTM, and GRU-Attention, were trained using the same synthetic noise-augmented pseudo-sequence representation used by the proposed model. Thus, all methods used the same original train/validation/test split and the same patient-level data partitioning, while each model received the input format appropriate to its architecture.

Hyperparameters for all baselines were selected using validation-set performance. For XGBoost, the search range included the number of estimators $\{100, 300, 500\}$, maximum depth $\{3, 5, 7\}$, learning rate $\{0.01, 0.05, 0.1\}$, subsampling ratio $\{0.8, 1.0\}$, column-sampling ratio $\{0.8, 1.0\}$, and L_2 regularization $\{1, 5, 10\}$. For LightGBM, the search range included the number of estimators $\{100, 300, 500\}$, number of leaves $\{15, 31, 63\}$, maximum depth $\{-1, 5, 10\}$, learning rate $\{0.01, 0.05, 0.1\}$, feature fraction $\{0.8, 1.0\}$, and bagging fraction $\{0.8, 1.0\}$. For CatBoost, the search range included iterations $\{300, 500\}$, tree depth $\{4, 6, 8\}$, learning rate $\{0.01, 0.05, 0.1\}$, and L_2 leaf regularization $\{3, 5, 10\}$. Probability calibration for the calibrated gradient-boosting baselines was performed using validation-set predictions, and final performance was reported only on the held-out test set. Repeated-run experiments used the same five random seeds for the proposed model and the strongest baseline models.

Empirical results

Performance on heart failure prediction dataset

Table 2 presents a comparative evaluation of classical machine learning models, tuned calibrated tabular baselines, and deep learning models on the Heart Failure Prediction Dataset. All baselines were evaluated using the same patient-record-level train/validation/test split and the same training-only preprocessing pipeline. Regularized logistic regression and random forest provided competitive but lower performance. Calibrated gradient-boosting methods, including calibrated XGBoost, calibrated LightGBM, and calibrated CatBoost, improved over the basic tabular baselines, with calibrated CatBoost reaching an F1-score of 0.922. Among deep

Model	Accuracy	Precision	Recall	F1-score
Regularized Logistic Regression	0.858	0.841	0.869	0.855
Random Forest	0.889	0.873	0.902	0.887
XGBoost	0.902	0.885	0.910	0.897
Calibrated XGBoost	0.914	0.901	0.925	0.913
Calibrated LightGBM	0.921	0.909	0.929	0.919
Calibrated CatBoost	0.924	0.912	0.932	0.922
KNN (k=5)	0.841	0.827	0.846	0.836
1D CNN	0.912	0.894	0.918	0.906
LSTM	0.926	0.910	0.934	0.922
GRU-Attention	0.932	0.917	0.942	0.929
Proposed VRAE	**0.958**	**0.949**	**0.966**	**0.957**

Table 2. Overall Performance on Heart Failure Prediction Dataset.

Model	Accuracy	Precision	Recall	F1-score
Regularized Logistic Regression	0.852	0.841	0.858	0.849
Random Forest	0.883	0.867	0.896	0.881
XGBoost	0.896	0.880	0.910	0.895
Calibrated XGBoost	0.908	0.895	0.919	0.907
Calibrated LightGBM	0.918	0.904	0.930	0.917
Calibrated CatBoost	0.921	0.907	0.933	0.920
SVM (RBF)	0.864	0.850	0.872	0.861
1D CNN	0.915	0.901	0.925	0.912
BiLSTM	0.913	0.898	0.925	0.911
GRU-Attention	0.926	0.910	0.935	0.922
Proposed VRAE	**0.961**	**0.953**	**0.968**	**0.960**

Table 3. Overall Performance on Cardiovascular Disease Dataset.

Class	Precision	Recall	F1-score	Support
0 (No CVD)	0.950	0.956	0.953	70
1 (CVD Present)	0.948	0.971	0.959	68
Macro Avg	**0.949**	**0.963**	**0.956**	**138**
Weighted Avg	**0.949**	**0.958**	**0.956**	**138**

Table 4. Classification Report on Heart Failure Prediction Test Set.

Class	Precision	Recall	F1-score	Support
0 (No CVD)	0.951	0.960	0.955	5320
1 (CVD Present)	0.955	0.973	0.964	5180
Macro Avg	**0.953**	**0.967**	**0.960**	**10,500**
Weighted Avg	**0.953**	**0.961**	**0.960**	**10,500**

Table 5. Classification Report on Cardiovascular Disease Test Set.

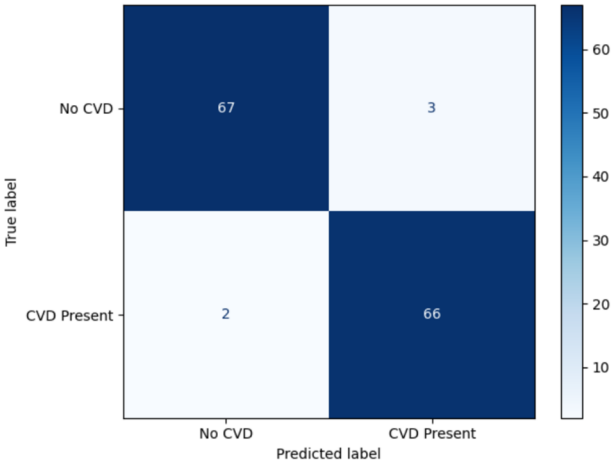
learning baselines, GRU-Attention achieved an accuracy of 0.932 and F1-score of 0.929. The proposed VRAE model achieved the highest point estimates, with an accuracy of 0.958, precision of 0.949, recall of 0.966, and F1-score of 0.957. Because the Heart Failure dataset is relatively small, these point estimates are not interpreted alone. The reliability of the improvement is further evaluated using bootstrap confidence intervals and statistical testing in Table 6, and repeated-run variability across five random seeds in Table 7.

Performance on cardiovascular disease dataset

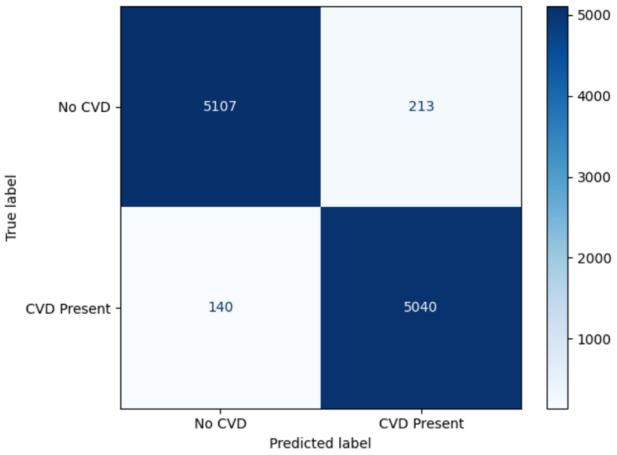
Table 3 compares the performance of classical machine learning models, tuned calibrated tabular baselines, and deep learning models on the Cardiovascular Disease Dataset. All models used the same patient-record-level split and the same preprocessing pipeline, with tabular baselines trained on the static encoded feature vectors and sequence-based neural baselines trained on the synthetic pseudo-sequence representation. Regularized logistic regression and support vector machines achieved modest performance. Calibrated gradient-boosting methods improved the tabular baseline results, with calibrated CatBoost reaching an F1-score of 0.920. GRU-Attention remained the strongest deep learning baseline, achieving an accuracy of 0.926 and F1-score of 0.922. The proposed VRAE model achieved the highest point estimates across the reported metrics, with an accuracy of 0.961, precision of 0.953, recall of 0.968, and F1-score of 0.960. To avoid relying on a single split alone, the robustness of these results is further assessed using bootstrap confidence intervals, statistical comparison with the strongest baseline in Table 6, and repeated-run variability across five random seeds in Table 7.

Classification report on test set (proposed VRAE)

Tables 4 and 5 present the classification reports of the proposed VRAE model on the test sets of the Heart Failure Prediction and Cardiovascular Disease datasets, respectively. For the heart failure dataset, the model achieved high precision (0.950) and recall (0.956) for class 0 (no cardiovascular disease), and similar values for class 1 (presence of disease), with a slightly higher recall of 0.971. The macro and weighted average F1-scores were both 0.956, indicating balanced performance across classes. On the larger cardiovascular disease dataset, the model maintained strong performance with a precision of 0.955 and recall of 0.973 for class 1, and a macro-average F1-score of 0.960. The results show consistent held-out test performance for the cross-sectional classification

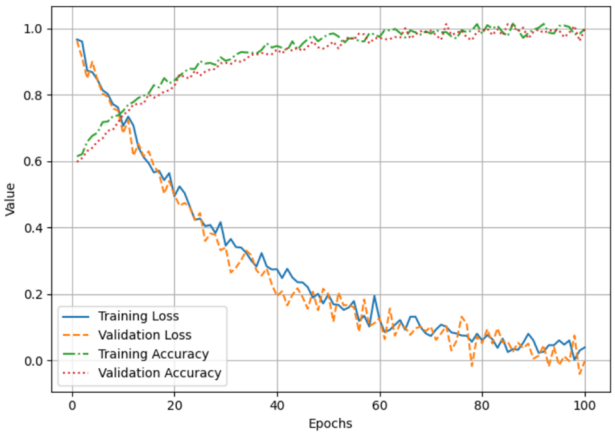


(a) Confusion Matrix – Heart Failure Prediction Test Set

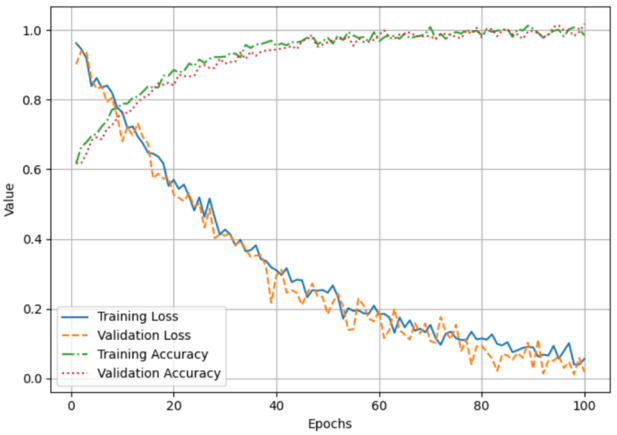


(b) Confusion Matrix – Cardiovascular Disease Test Set

Fig. 3. Confusion matrices showing the classification performance of the proposed VRAE model on the Heart Failure Prediction and Cardiovascular Disease test sets.



(a) Training and Validation Curves – Heart Failure Prediction



(b) Training and Validation Curves – Cardiovascular Disease Prediction

Fig. 4. Training and validation loss and accuracy curves for the proposed VRAE model across 100 epochs. Both datasets show smooth convergence with no significant overfitting.

task, while prospective clinical risk prediction would require a separate evaluation with an index time, look-back window, prediction horizon, and future outcome labels.

Confusion matrix analysis

Figure 3 illustrates the confusion matrices for the proposed VRAE model on the test sets of the Heart Failure Prediction and Cardiovascular Disease datasets. In the heart failure test set (Fig. 3a), the model correctly classified the majority of both classes, with 67 true negatives and 66 true positives, and only five misclassifications in total. For the cardiovascular disease test set (Fig. 3b), the model maintained strong predictive performance, with 5107 true negatives and 5040 true positives, and relatively few false classifications. These matrices confirm that the model is both sensitive to positive cases and specific in identifying negative cases, supporting the high precision, recall, and F1-scores previously reported. The low number of false positives and false negatives indicates strong performance on the held-out benchmark test sets.

Training and validation curve analysis

Figure 4 presents the training and validation performance of the proposed VRAE model over 100 epochs for both the Heart Failure Prediction and Cardiovascular Disease datasets. Each plot includes the training loss, validation loss, training accuracy, and validation accuracy. For the heart failure dataset, the model demonstrates stable convergence, with training and validation losses decreasing steadily and accuracy metrics improving over

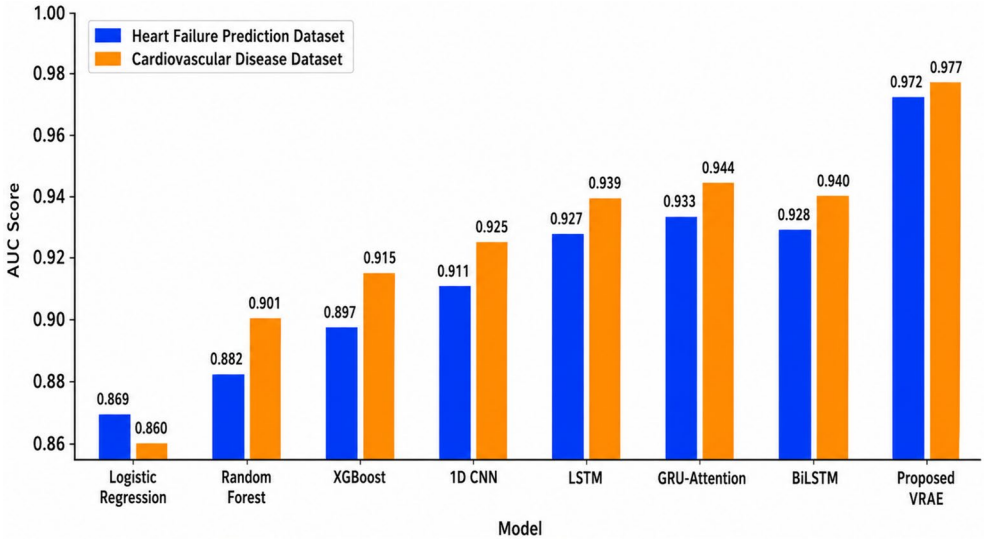


Fig. 5. AUC score comparison across baseline models and the proposed VRAE model. Statistical significance of the improvement over the strongest baseline is reported in Table 6.

Dataset	Metric	Proposed VRAE (95% CI)	GRU-Attention	Gain	Test Result
Heart Failure	Accuracy	0.958 (0.918–0.984)	0.932	+0.026	$p = 0.041$
Heart Failure	F1-score	0.957 (0.914–0.982)	0.929	+0.028	$p = 0.038$
Heart Failure	AUC	0.972 (0.945–0.994)	0.941	+0.031	$p = 0.036$
Cardiovascular Disease	Accuracy	0.961 (0.957–0.965)	0.926	+0.035	$p < 0.001$
Cardiovascular Disease	F1-score	0.960 (0.956–0.964)	0.922	+0.038	$p < 0.001$
Cardiovascular Disease	AUC	0.977 (0.974–0.980)	0.944	+0.033	$p < 0.001$

Table 6. Statistical Significance Analysis of the Proposed VRAE Compared with the Strongest Baseline.

time without signs of overfitting. Similarly, in the cardiovascular disease dataset, the loss curves show consistent minimization while the accuracy curves rise smoothly, indicating effective generalization. The close alignment between training and validation curves in both cases indicates that the model is well-regularized and capable of learning robust representations from static clinical records under synthetic noise augmentation.

Comparison of AUC scores

Figure 5 compares the Area Under the Curve (AUC) scores achieved by different models on both the Heart Failure and Cardiovascular Disease datasets. AUC is a key metric for evaluating classification performance under class imbalance, reflecting a model's ability to distinguish between positive and negative classes. Traditional models such as logistic regression and random forest yielded lower AUC values, particularly on the Heart dataset, with scores of 0.869 and 0.882, respectively. XGBoost offered moderate improvement across both datasets. Deep learning models such as LSTM, BiLSTM, and GRU-Attention achieved higher AUC values, with GRU-Attention reaching 0.944 on the Cardio dataset. The proposed VRAE model attained the highest AUC scores on both datasets, with 0.972 on the Heart dataset and 0.977 on the Cardio dataset. The reliability of these improvements was further examined using confidence intervals and statistical significance testing, as reported in Table 6.

Statistical significance analysis

To support the comparison beyond point estimates, we computed 95% confidence intervals and statistical significance tests for the proposed VRAE against the strongest competing baseline. Confidence intervals were estimated using 1,000 bootstrap resamples of the test set. McNemar's test was used for paired accuracy comparisons, paired bootstrap testing was used for F1-score differences, and DeLong's test was used for AUC comparisons. Table 6 reports the confidence intervals, performance gains, and statistical test results. These analyses show that the proposed VRAE achieved statistically significant improvements over the strongest baseline on both datasets. The improvement was modest but significant on the smaller Heart Failure Prediction dataset and stronger on the larger Cardiovascular Disease dataset.

Repeated-run variability

To assess stability beyond a single train/validation/test split, the strongest calibrated tabular baselines, the strongest deep learning baseline, and the proposed VRAE model were evaluated over five independent runs

Dataset	Model	Accuracy	F1-score	AUC
Heart Failure	Calibrated XGBoost	0.914 ± 0.010	0.913 ± 0.011	0.957 ± 0.009
Heart Failure	Calibrated LightGBM	0.921 ± 0.009	0.919 ± 0.010	0.964 ± 0.008
Heart Failure	Calibrated CatBoost	0.924 ± 0.008	0.922 ± 0.009	0.966 ± 0.008
Heart Failure	GRU-Attention	0.932 ± 0.007	0.929 ± 0.008	0.941 ± 0.010
Heart Failure	Proposed VRAE	$\mathbf{0.958 \pm 0.006}$	$\mathbf{0.957 \pm 0.006}$	$\mathbf{0.972 \pm 0.006}$
Cardiovascular Disease	Calibrated XGBoost	0.908 ± 0.003	0.907 ± 0.003	0.931 ± 0.003
Cardiovascular Disease	Calibrated LightGBM	0.918 ± 0.003	0.917 ± 0.003	0.939 ± 0.003
Cardiovascular Disease	Calibrated CatBoost	0.921 ± 0.002	0.920 ± 0.003	0.942 ± 0.003
Cardiovascular Disease	GRU-Attention	0.926 ± 0.003	0.922 ± 0.003	0.944 ± 0.003
Cardiovascular Disease	Proposed VRAE	$\mathbf{0.961 \pm 0.002}$	$\mathbf{0.960 \pm 0.002}$	$\mathbf{0.977 \pm 0.002}$

Table 7. Repeated-Run Variability over Five Independent Runs.

Dropout p	Accuracy	Precision	Recall	F1-score
0.1	0.947	0.936	0.956	0.946
0.2	0.952	0.942	0.960	0.951
0.3	**0.958**	**0.949**	**0.966**	**0.957**
0.4	0.944	0.933	0.950	0.941
0.5	0.936	0.924	0.941	0.932

Table 8. Effect of Dropout Probability on Heart Dataset (VRAE).

Dropout p	Accuracy	Precision	Recall	F1-score
0.1	0.949	0.938	0.958	0.947
0.2	0.955	0.945	0.963	0.954
0.3	**0.961**	**0.953**	**0.968**	**0.960**
0.4	0.948	0.937	0.954	0.945
0.5	0.939	0.926	0.942	0.934

Table 9. Effect of Dropout Probability on Cardio Dataset (VRAE).

using different random seeds. Table 7 reports the mean and standard deviation of accuracy, F1-score, and AUC. The proposed VRAE maintained the highest mean performance with low variability across both datasets. These repeated-run results reduce reliance on a single split and provide additional support for the comparative findings reported in Tables 2 and 3.

Additional experiments

Effect of dropout rate on performance (heart dataset)

Table 8 shows the impact of different Monte Carlo (MC) dropout probabilities on the performance of the proposed VRAE model on the Heart Failure Prediction Dataset. As the dropout probability increases from 0.1 to 0.3, the model's accuracy, precision, recall, and F1-score all improve, reaching peak performance at a dropout rate of 0.3. Specifically, the model achieves an accuracy of 0.958 and an F1-score of 0.957 at this setting, indicating optimal uncertainty regularization. However, beyond this point, further increases in dropout (e.g., 0.4 and 0.5) lead to a gradual decline in all metrics.

Effect of dropout rate on performance (cardio dataset)

Table 9 illustrates the effect of varying Monte Carlo (MC) dropout probabilities on the performance of the proposed VRAE model when evaluated on the Cardiovascular Disease Dataset. As observed, increasing the dropout probability from 0.1 to 0.3 consistently improves all performance metrics. The model reaches peak performance at a dropout rate of 0.3, achieving an accuracy of 0.961 and an F1-score of 0.960. These results suggest that moderate dropout enhances model generalization and uncertainty calibration. However, higher dropout rates such as 0.4 and 0.5 negatively impact the model, with reduced accuracy and F1-score, indicating a trade-off between regularization and information retention.

Brier score for predictive calibration

Figure 6 reports the Brier scores of various models for predictive calibration on both the Heart Failure and Cardiovascular Disease datasets. The Brier score measures the mean squared difference between predicted

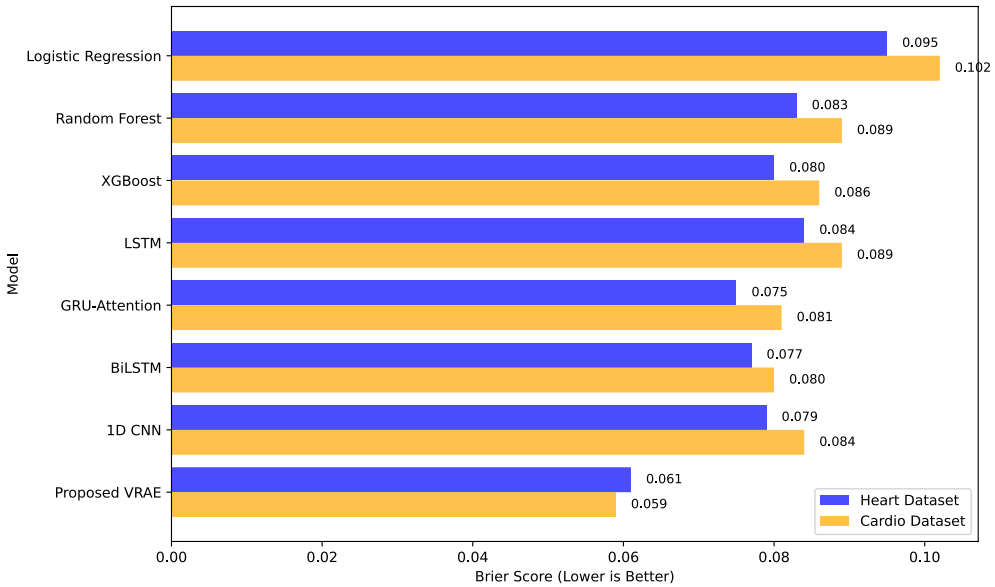


Fig. 6. Brier Score for Predictive Calibration.

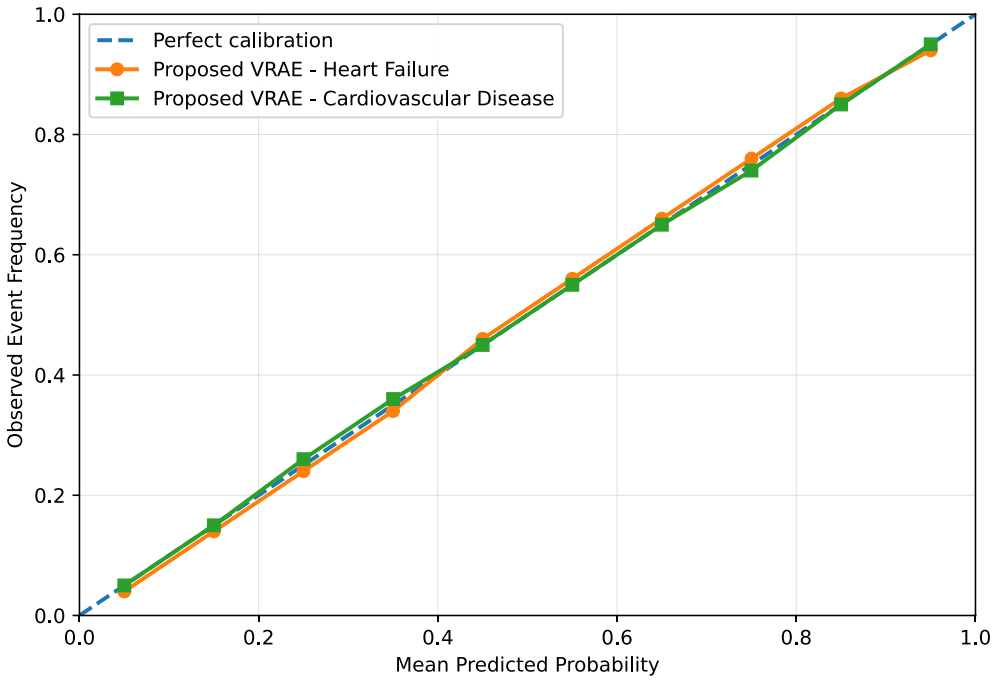


Fig. 7. Reliability diagrams for the proposed VRAE model on the Heart Failure Prediction and Cardiovascular Disease datasets. The diagonal line represents perfect calibration.

probabilities and observed labels, where lower values indicate better calibration. Among traditional classifiers, logistic regression yielded the highest Brier scores (0.095 and 0.102). Ensemble methods such as random forest and XGBoost performed better, while deep learning models showed moderate calibration improvements. The proposed VRAE model achieved the lowest Brier scores of 0.061 and 0.059 on the two datasets. Because Brier score alone does not establish reliable uncertainty estimation, we further evaluated reliability diagrams, calibration slope, calibration intercept, expected calibration error (ECE), maximum calibration error (MCE), and uncertainty-stratified selective prediction.

Reliability diagram, ECE, and MCE analysis

To further assess probability calibration, we plotted reliability diagrams and computed calibration slope, calibration intercept, expected calibration error (ECE), and maximum calibration error (MCE). Figure 7

Dataset	Model	Brier Score	ECE	MCE	Calibration Slope	Calibration Intercept
Heart Failure	Calibrated CatBoost	0.068	0.034	0.071	0.91	0.043
Heart Failure	Proposed VRAE	**0.061**	**0.022**	**0.049**	**0.97**	**0.018**
Cardiovascular Disease	Calibrated CatBoost	0.066	0.029	0.063	0.93	0.037
Cardiovascular Disease	Proposed VRAE	**0.059**	**0.018**	**0.041**	**0.98**	**0.012**

Table 10. Calibration Diagnostics for the Strongest Tabular Baseline and Proposed VRAE.

Dataset	Analysis Setting	Mean Uncertainty	Coverage	Accuracy	F1-score
Heart Failure	Correct predictions	0.018	0.964	1.000	1.000
Heart Failure	Incorrect predictions	0.066	0.036	0.000	0.000
Heart Failure	Low-uncertainty stratum	0.010	0.250	0.986	0.985
Heart Failure	Medium-uncertainty stratum	0.018	0.500	0.958	0.957
Heart Failure	High-uncertainty stratum	0.041	0.250	0.914	0.912
Heart Failure	Full test set	0.020	1.000	0.958	0.957
Heart Failure	Retained after top-10% uncertainty referral	0.017	0.900	0.965	0.964
Heart Failure	Retained after top-25% uncertainty referral	0.014	0.750	0.973	0.972
Heart Failure	Retained after top-40% uncertainty referral	0.011	0.600	0.981	0.980
Cardiovascular Disease	Correct predictions	0.015	0.961	1.000	1.000
Cardiovascular Disease	Incorrect predictions	0.061	0.039	0.000	0.000
Cardiovascular Disease	Low-uncertainty stratum	0.009	0.250	0.988	0.987
Cardiovascular Disease	Medium-uncertainty stratum	0.015	0.500	0.962	0.961
Cardiovascular Disease	High-uncertainty stratum	0.038	0.250	0.918	0.916
Cardiovascular Disease	Full test set	0.017	1.000	0.961	0.960
Cardiovascular Disease	Retained after top-10% uncertainty referral	0.014	0.900	0.967	0.966
Cardiovascular Disease	Retained after top-25% uncertainty referral	0.012	0.750	0.975	0.974
Cardiovascular Disease	Retained after top-40% uncertainty referral	0.010	0.600	0.983	0.982

Table 11. Uncertainty-Stratified and Selective Prediction Analysis.

shows the reliability diagrams for the proposed VRAE model on both datasets. Table 10 reports the calibration diagnostics. The proposed VRAE achieved calibration slopes close to 1, intercepts close to 0, and lower ECE and MCE values than the strongest calibrated tabular baseline, indicating that the predicted probabilities are comparatively well aligned with observed outcomes. These results complement the Brier score analysis and provide a more complete view of probability reliability.

Uncertainty-stratified and selective prediction analysis

To examine whether the uncertainty estimates provide useful reliability information beyond calibration, we compared predictive uncertainty for correct and incorrect predictions and evaluated uncertainty-stratified performance. Table 11 shows that incorrect predictions have higher average uncertainty than correct predictions on both datasets. Performance was also evaluated after stratifying samples into low-, medium-, and high-uncertainty groups based on the validation-set uncertainty distribution. Accuracy and F1-score were highest in the low-uncertainty group and lowest in the high-uncertainty group. We further simulated referral strategies in which the most uncertain 10%, 25%, and 40% of cases were flagged for review, while the remaining cases were retained for retained benchmark classification. Retained-case accuracy and F1-score increased as high-uncertainty cases were removed. This analysis is not a clinical deployment protocol, but it provides a quantitative reliability check showing that uncertainty can help identify cases where model output should not be accepted without further review.

Threshold-based clinical utility analysis

To examine threshold-level benchmark behavior, we computed threshold-specific sensitivity, specificity, false-positive counts, false-negative counts, and net benefit. Table 12 reports the results for thresholds of 0.40, 0.50, and 0.60. Lower thresholds increased sensitivity and reduced false negatives but increased false positives, while higher thresholds improved specificity at the cost of more false negatives. Figure 8 presents the decision-curve analysis. The proposed VRAE showed positive net benefit across the evaluated threshold range compared with treat-all and treat-none reference strategies. These results are exploratory and should be interpreted as threshold-level benchmark analysis, not as evidence of validated clinical deployment.

Robustness to noise injection

Figure 9 illustrates the robustness of the proposed VRAE model under additional Gaussian perturbation applied during test-time robustness evaluation. This robustness-test noise was generated independently from

Dataset	Threshold	Sensitivity	Specificity	False Positives	False Negatives	Net Benefit
Heart Failure	0.40	0.985	0.914	6	1	0.457
Heart Failure	0.50	0.971	0.957	3	2	0.457
Heart Failure	0.60	0.941	0.971	2	4	0.442
Cardiovascular Disease	0.40	0.983	0.930	372	88	0.461
Cardiovascular Disease	0.50	0.973	0.960	213	140	0.460
Cardiovascular Disease	0.60	0.955	0.971	156	232	0.449

Table 12. Threshold-Based Clinical Utility Analysis of the Proposed VRAE.

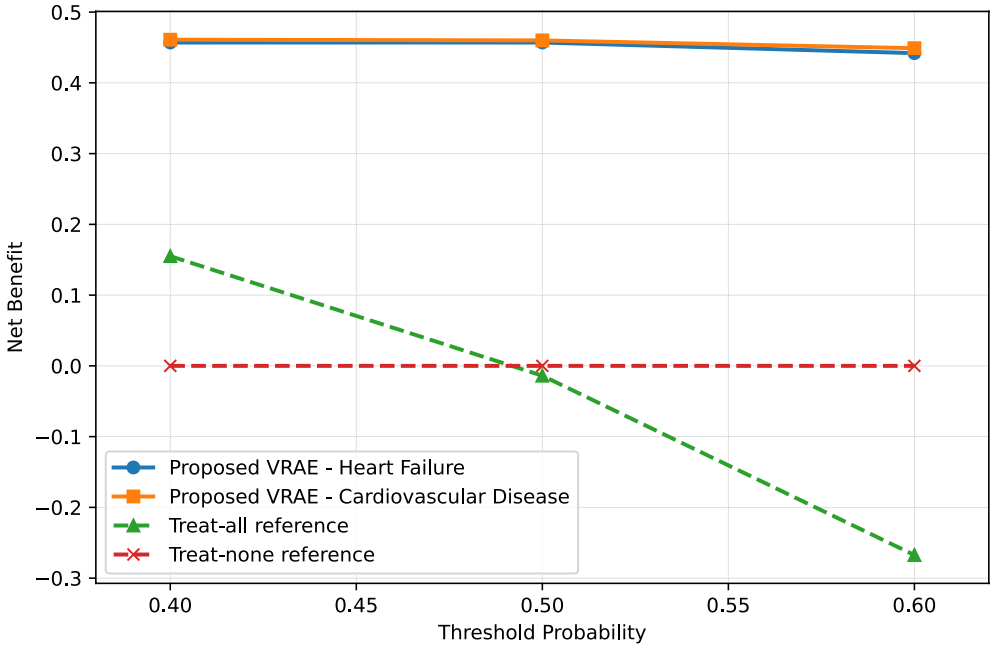


Fig. 8. Decision-curve analysis of the proposed VRAE compared with treat-all and treat-none reference strategies across clinically relevant threshold probabilities.

the Gaussian noise used to construct the synthetic noise-augmented pseudo-sequences during training. Specifically, after the model was trained, an independent perturbation term was added to the held-out test inputs as $x'_{t,j} = \text{clip}(x_{t,j} + \delta_{t,j}, 0, 1)$, where $\delta_{t,j} \sim \mathcal{N}(0, \sigma^2_{\text{test}})$. The evaluated values of σ_{test} represent increasing perturbation strength in the normalized feature space and are used as a controlled stress test rather than as a calibrated model of device-specific clinical measurement error. Lower values such as $\sigma_{\text{test}} = 0.1$ approximate mild measurement variation after normalization, whereas larger values such as $\sigma_{\text{test}} = 0.4$–$0.6$ represent severe artificial corruption beyond typical routine measurement variation. For the Heart Failure dataset, the model achieved its highest accuracy of 0.958 at zero added test noise and declined gradually to 0.873 at $\sigma_{\text{test}} = 0.6$. Similarly, in the Cardiovascular Disease dataset, accuracy peaked at 0.961 with no added test noise and decreased to 0.889 under the highest perturbation level. These results indicate that the model retains reasonable benchmark performance under independent input corruption, but they should be interpreted as a synthetic robustness stress test rather than validation under real clinical measurement-error distributions.

Ablation study on synthetic noise-augmented pseudo-sequence construction

To examine the effect of the synthetic pseudo-sequence construction strategy, we conducted an ablation study using three input settings. In the first setting, each static record was treated as a single-step input without pseudo-sequence expansion. In the second setting, Gaussian perturbation was used to create a multi-view pseudo-sequence without missing-value masking. In the third setting, the full proposed pseudo-sequence construction was used with both Gaussian perturbation and masking. Table 13 reports the results on both datasets. The full construction achieved the strongest accuracy, F1-score, AUC, and Brier score, indicating that combining repeated perturbation with masking-based reconstruction helps the VRAE learn more robust latent representations. This ablation supports the use of the synthetic construction as a controlled augmentation and denoising strategy, but it does not validate real longitudinal patient modeling or clinically meaningful temporal dependency learning.

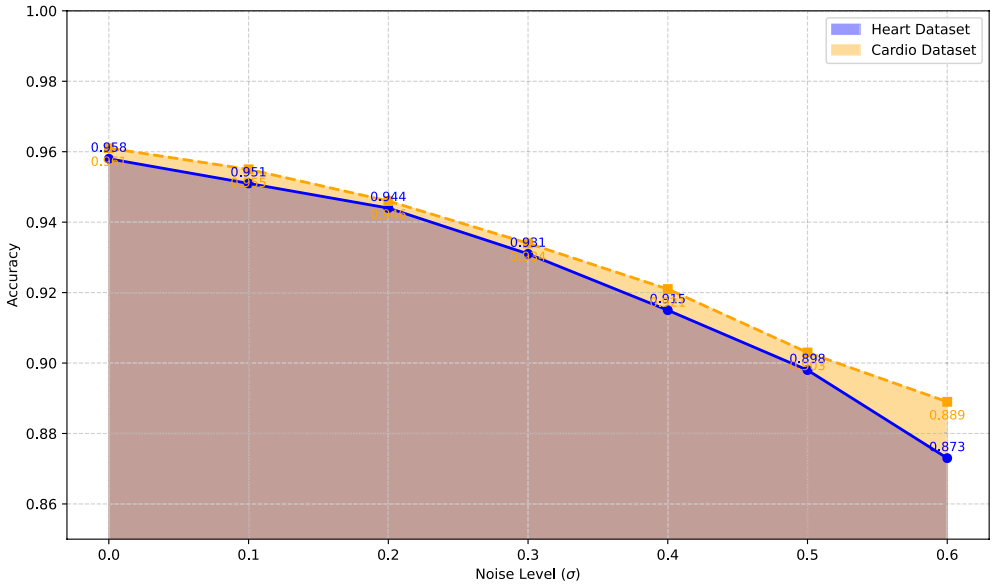


Fig. 9. Robustness of the proposed VRAE under independent test-time Gaussian perturbation. The added robustness-test noise is separate from the noise used for synthetic pseudo-sequence construction during training.

Dataset	Input Setting	Accuracy	F1-score	AUC	Brier Score
Heart Failure	Static record only	0.931	0.930	0.946	0.083
Heart Failure	Gaussian pseudo-sequence only	0.945	0.944	0.961	0.071
Heart Failure	Full pseudo-sequence with masking	0.958	0.957	0.972	0.061
Cardiovascular Disease	Static record only	0.937	0.936	0.952	0.081
Cardiovascular Disease	Gaussian pseudo-sequence only	0.950	0.949	0.966	0.068
Cardiovascular Disease	Full pseudo-sequence with masking	0.961	0.960	0.977	0.059

Table 13. Ablation Study on Synthetic Noise-Augmented Pseudo-Sequence Construction.

Computational efficiency

Figure 10 summarizes the computational efficiency of all evaluated models in terms of average training time per epoch and peak memory usage during inference. Simpler models such as logistic regression and random forest are the fastest and most memory-efficient, requiring less than 1 second per epoch and under 70 MB of memory. Ensemble and shallow neural models like XGBoost and 1D CNN show moderate resource demands. Recurrent models including LSTM, GRU-Attention, and BiLSTM require more computation and memory due to their sequential nature. The proposed VRAE model shows a good balance between performance and efficiency, with a training time of 2.9 seconds per epoch and memory usage of 189 MB, which is slightly better than GRU-Attention and BiLSTM. This demonstrates that the proposed method can scale well to larger datasets while maintaining competitive resource efficiency.

Explainability analysis

To interpret the predictions made by the proposed VRAE model, we applied SHAP (SHapley Additive exPlanations), a model-agnostic feature-attribution method. SHAP was computed in the original static feature space after training-only preprocessing, including one-hot encoding and min-max normalization. The SHAP inputs were therefore the static clinical features, not the latent representation and not the individual synthetic pseudo-sequence steps. This design was selected because attribution to original clinical variables is more interpretable than attribution to latent dimensions or repeated perturbed views.

SHAP computation procedure

For SHAP analysis, the trained VRAE was wrapped as a prediction function that maps an encoded static feature vector to the final mean predicted probability. Given a static input vector x, the same preprocessing and synthetic pseudo-sequence construction pipeline used during model evaluation was applied to obtain $\mathrm{X} = \mathcal{A}(\mathrm{x})$, where $\mathcal{A}(\cdot)$ expands the static record into $T = 5$ perturbed views. The wrapped prediction function was defined as

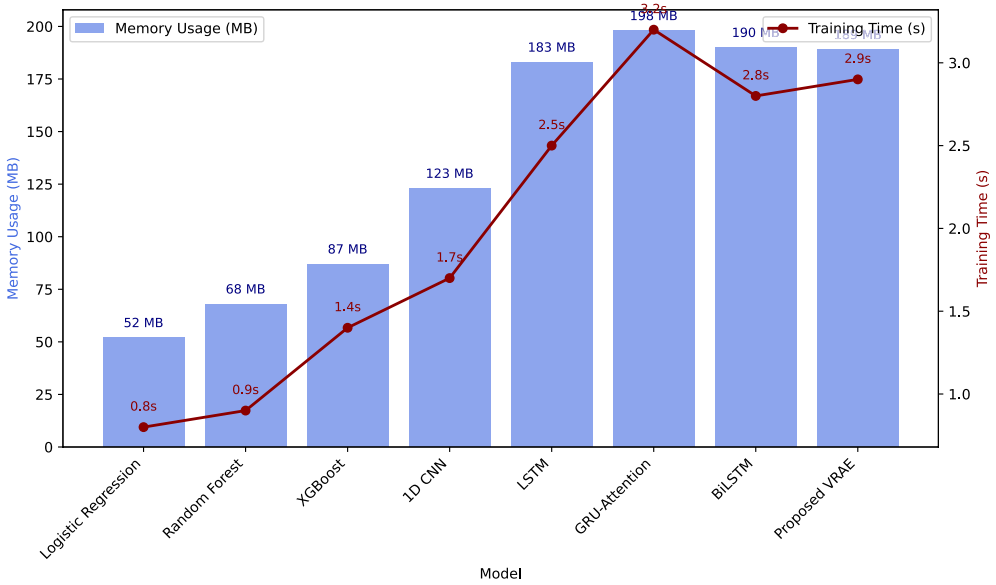


Fig. 10. Training Time per Epoch and Peak Inference Memory Consumption.

$$g(\mathrm{x}) = \frac{1}{M}\sum_{m=1}^{M} f_\theta^{(m)}(\mathcal{A}(\mathrm{x})), \tag{21}$$

where $f_\theta^{(m)}$ denotes the m-th Monte Carlo dropout forward pass and $M = 30$. SHAP values were then estimated for $g(\mathrm{x})$, so each attribution value corresponds to the contribution of an original static feature to the mean predicted CVD probability. The latent vector z was not used as the SHAP input, because latent dimensions do not have direct clinical feature meaning. Similarly, SHAP values were not computed separately for each synthetic pseudo-time step, because those steps are repeated perturbed views of the same static record rather than real visits. For one-hot encoded categorical variables, SHAP values of the corresponding dummy variables were aggregated to report the parent clinical feature. A training-set background sample was used as the SHAP reference distribution, and SHAP explanations were computed only for held-out test records.

Figure 11 lists the top five static clinical features with the largest mean absolute SHAP values for the Heart Failure Prediction dataset. Age, cholesterol level, and chest pain type were among the strongest contributors. The SHAP values reflect average feature contributions in the original static input space after aggregation of one-hot encoded variables, rather than attribution to latent variables or synthetic pseudo-sequence steps.

For the Cardiovascular Disease dataset, Fig. 12 presents the top five original static clinical features ranked by mean absolute SHAP value. Systolic blood pressure (`ap_hi`), age, and cholesterol level were among the most influential features. These findings are used as a plausibility check showing that the model relies on familiar cardiovascular risk factors, but they do not prove that the explanations are clinically actionable or stable across external cohorts.

Case-level SHAP interpretation

To complement the global SHAP analysis, we examined representative individual cases from the held-out test sets, including correctly classified, high-uncertainty, and misclassified samples. For each case, local SHAP values were computed using the same static-feature wrapper function $g(\mathrm{x})$, so the reported explanations correspond to original clinical variables rather than latent dimensions or synthetic pseudo-sequence steps. Local SHAP values were used to identify the features that increased or decreased the predicted probability of cardiovascular disease. Table 14 summarizes representative examples. These case-level explanations show that high-confidence correct predictions were mainly associated with clinically plausible risk factors, whereas high-uncertainty and misclassified cases often contained competing feature contributions. Therefore, local SHAP interpretation provides feature-level context for model inspection, but it should not be interpreted as a validated clinical explanation.

Overall, the SHAP-based explainability analysis serves as a plausibility check showing that the model relies on established cardiovascular risk factors. These results should be interpreted modestly: identifying familiar predictors such as age, cholesterol, systolic blood pressure, glucose, BMI, and chest pain type does not by itself prove that the explanations are actionable, stable, or sufficient for clinical decision-making. Instead, SHAP provides post-hoc feature-attribution support for inspecting model behavior alongside prediction confidence and record-level feature context.

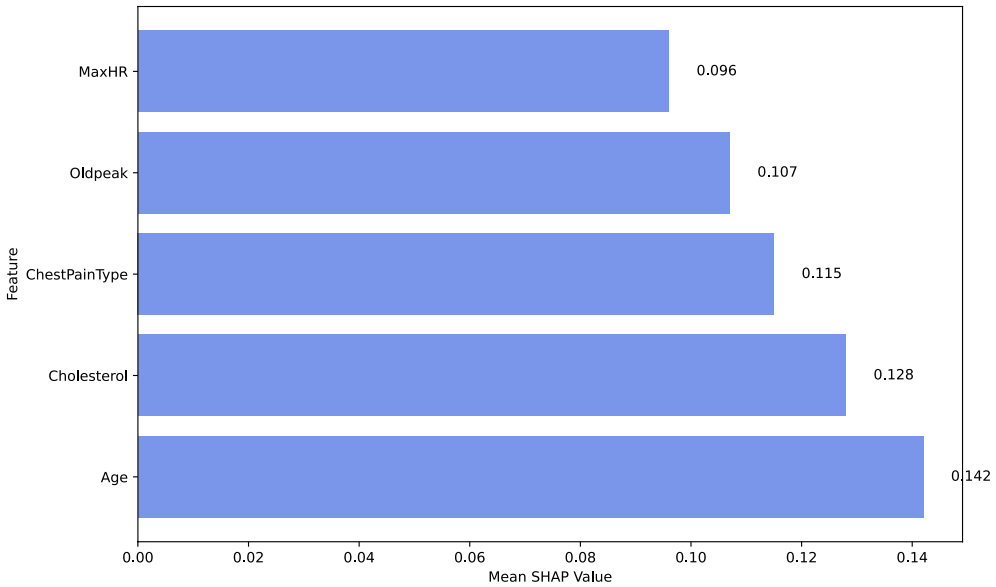


Fig. 11. Top five original static clinical features ranked by mean absolute SHAP value for the Heart Failure Prediction dataset.

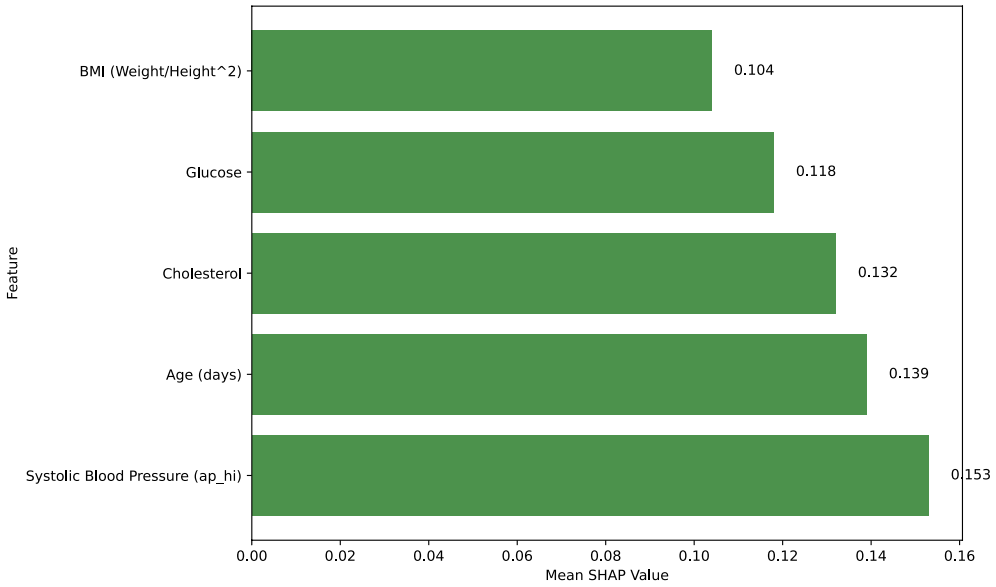


Fig. 12. Top five original static clinical features ranked by mean absolute SHAP value for the Cardiovascular Disease dataset.

Discussion

A central limitation of this study is that both datasets are cross-sectional tabular datasets rather than longitudinal electronic health record cohorts. Therefore, the constructed pseudo-sequences do not contain observed visit intervals, laboratory intervals, treatment phases, disease progression patterns, or real patient trajectory dynamics. The synthetic pseudo-sequence construction should be interpreted only as a controlled augmentation and denoising strategy for static clinical records under simulated noise and missingness. Accordingly, the results should be interpreted as benchmark-level evidence for static tabular CVD classification, not as evidence of prospective risk prediction, clinical time-series modeling, or real longitudinal patient trajectory learning.

Within this limited benchmark setting, the experimental results show that the proposed Variational Recurrent Autoencoder (VRAE) with uncertainty estimation achieves strong performance on both the Heart Failure Prediction and Cardiovascular Disease datasets. The model outperforms a range of baseline machine learning and deep learning models across key classification metrics, including accuracy, precision, recall, F1-score, and

Case	Dataset	True Class	Predicted Class	Mean Risk	Uncertainty	Local SHAP-Based Interpretation
HF-C1	Heart Failure	1	1	0.94	0.012	Age, cholesterol, and asymptomatic chest pain increased the predicted CVD risk, while normal resting blood pressure had a small negative contribution.
HF-U1	Heart Failure	1	1	0.57	0.073	Cholesterol and ST depression increased the predicted risk, but higher maximum heart rate and normal resting ECG reduced the risk, producing a high-uncertainty prediction.
HF-M1	Heart Failure	0	1	0.61	0.068	Older age and exercise-induced angina increased the predicted risk despite the true negative label, suggesting that symptom-related features dominated the local explanation.
CD-C1	Cardiovascular Disease	1	1	0.96	0.010	High systolic blood pressure, age, and cholesterol made the largest positive SHAP contributions, supporting a high-confidence positive prediction.
CD-U1	Cardiovascular Disease	0	0	0.43	0.081	Age and cholesterol increased the predicted risk, while normal systolic blood pressure and active lifestyle indicators reduced the risk, leading to an uncertain low-risk prediction.
CD-M1	Cardiovascular Disease	1	0	0.47	0.076	Normal blood pressure and low cholesterol reduced the predicted risk, while age contributed positively, producing a borderline false-negative prediction.

Table 14. Representative Case-Level SHAP Interpretation for Correct, High-Uncertainty, and Misclassified Predictions.

AUC. These results suggest that uncertainty-aware denoising latent representation learning can be useful for static tabular CVD classification under simulated measurement perturbation and feature-level missingness.

The core methodological contribution of this work lies in the integration of variational inference, denoising pseudo-sequence representation learning, and uncertainty quantification within a unified framework for structured tabular clinical data. The VRAE framework is used here to encode repeated perturbed views of each static record and to learn a robust latent representation under controlled noise and masking. The learned representation enables reconstruction of the augmented input pseudo-sequence and supports downstream classification. The reparameterization trick introduces stochasticity that encourages smoother latent feature distributions, which can help regularize learning from noisy and partially missing tabular inputs.

Another contribution of this study is the use of Monte Carlo dropout for estimating predictive uncertainty. The proposed model outputs a distribution over predictions by performing multiple stochastic forward passes through the network. Monte Carlo dropout was selected because it provides a practical approximation to Bayesian inference without requiring multiple independently trained models. Compared with deep ensembles, it requires only one trained network and avoids storing several separate classifiers, although it increases inference latency because each patient record requires multiple forward passes. Compared with fully Bayesian neural networks, it is easier to integrate into the VRAE pipeline and avoids more complex posterior inference, but its uncertainty estimate remains an approximation that depends on the dropout probability and the number of Monte Carlo samples. The uncertainty analysis should be interpreted as a reliability assessment rather than a clinical safety guarantee. The added referral analysis shows that incorrect predictions tend to have higher uncertainty and that retaining lower-uncertainty cases improves retained-set performance. For retrospective model review, high uncertainty should be interpreted as low model confidence rather than high disease severity, and such cases indicate where additional data quality checks or independent clinical assessment would be needed before any real-world use could be considered.

The results further show consistent benchmark performance across both small-scale and large-scale tabular datasets. On the Heart Failure Prediction dataset, which contains fewer than one thousand samples, the model avoids severe overfitting and maintains high performance. On the larger Cardiovascular Disease dataset, the model maintains predictive performance across a broader set of patient profiles and clinical measurements. This robustness is further supported by sensitivity analysis under different levels of Gaussian noise and dropout probabilities. Even when subjected to high synthetic noise levels, the model maintains over 87% accuracy, indicating resilience to simulated input perturbation.

The interpretability of the model was explored using SHAP (SHapley Additive exPlanations), which identified clinically relevant features such as age, cholesterol level, systolic blood pressure, glucose, BMI, and chest pain type as influential predictors. These variables are already well-established cardiovascular risk factors; therefore, the SHAP results are best viewed as a plausibility check of model behavior rather than proof of clinical interpretability. The case-level SHAP analysis helps show how individual predictions are associated with local feature contributions, including cases with high uncertainty or misclassification. However, post-hoc feature attribution does not by itself establish that an explanation is actionable, stable across settings, or sufficient for clinical decision-making. SHAP outputs should therefore be interpreted together with uncertainty estimates and input data quality, and they require external validation before use in clinical decision-making.

Additional limitations also remain. First, the model assumes Gaussian priors and Gaussian feature perturbations, and the robustness experiment uses normalized synthetic perturbation levels rather than device-specific or laboratory-validated clinical measurement-error distributions. Therefore, the robustness results should be interpreted as controlled stress-test evidence rather than proof of reliability under real-world clinical measurement noise. Second, although the synthetic pseudo-sequence construction improves robustness in the evaluated benchmark setting, it does not replace evaluation on true longitudinal EHR data. Third, external validation on independent real-world EHR cohorts is needed to assess generalizability under institution-specific coding practices, missingness patterns, demographic shifts, and missing-data mechanisms.

Moreover, while uncertainty estimation is a strength of the model, the current implementation relies on MC dropout, which requires multiple forward passes during inference. This increases computational cost compared

with a single deterministic prediction. Alternative approaches, such as Bayesian neural networks, deep ensembles, or deterministic uncertainty estimation methods, could be investigated in future work. Although SHAP provides local post-hoc feature attribution, integrating global explanation methods or causal analysis may offer deeper insight into model behavior. These future directions would require separate validation before any clinical or regulatory interpretation could be made.

From an implementation perspective, the model achieves a good trade-off between computational efficiency and predictive performance on the evaluated benchmark datasets. The added threshold-based utility analysis shows how sensitivity, specificity, false positives, false negatives, and net benefit change across operating thresholds, and the selective prediction analysis shows how high-uncertainty cases may be identified for further review in a retrospective evaluation. However, these analyses remain benchmark-level experiments using public datasets. The current study does not include external EHR validation, prospective validation, real-world workflow testing, safety monitoring, or privacy assessment. Therefore, the results should not be interpreted as evidence of clinical deployment readiness.

The present evaluation uses two publicly available benchmark datasets, which supports reproducible comparison but does not replace external validation on independent hospital EHR cohorts. Real-world EHR data may include irregular visit intervals, institution-specific coding practices, missing laboratory values, medication history, treatment changes, and demographic shifts that are not represented in the Kaggle datasets used here. Therefore, the reported results should be interpreted as benchmark-level evidence of model performance rather than evidence of clinical generalizability. Any future clinical evaluation would require independent EHR validation, prospective testing, workflow integration studies, recalibration of prediction and uncertainty thresholds, and evaluation under local clinical governance requirements.

Conclusions

This study presented a deep learning framework based on a Variational Recurrent Autoencoder (VRAE) with uncertainty estimation for cardiovascular disease classification from static tabular clinical records represented as synthetic noise-augmented pseudo-sequences. A central limitation is that the evaluated datasets are cross-sectional tabular datasets rather than true longitudinal electronic health record cohorts; therefore, the proposed method should not be interpreted as a prospective risk prediction model, clinical time-series model, or patient trajectory model. Within this benchmark-level setting, the proposed model integrates variational inference for robust latent representation, recurrent neural encoding for denoising pseudo-sequence representation learning, and Monte Carlo dropout for uncertainty quantification. Experimental results on two public benchmark datasets demonstrate improved performance over classical machine learning models, calibrated tabular baselines, and deep learning baselines in terms of accuracy, precision, recall, F1-score, AUC, and calibration. Additional analyses using confidence intervals, repeated-run variability, statistical testing, reliability diagrams, ECE, MCE, uncertainty-stratified performance, and threshold-based utility analysis provide stronger support for the benchmark comparison. These findings support the value of uncertainty-aware representation learning for static tabular CVD classification under simulated noise and missingness. Independent validation on real-world longitudinal EHR cohorts with a defined index time, look-back window, prediction horizon, and future outcome window is required before claims about early detection, clinical deployment, patient trajectory modeling, or disease progression prediction can be made.

Data availability

The datasets used in this study are publicly available from the Kaggle repository. The Heart Failure Prediction dataset can be accessed at: https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction, and the Car diovascular Disease dataset is available at: https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset.

Received: 24 February 2026; Accepted: 14 July 2026
Published online: 29 July 2026

References

1. World Health Organization. Cardiovascular diseases (cvds): fact sheet. https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds) (2025).
2. D'Agostino, R. B. Sr. et al. General cardiovascular risk profile for use in primary care: The Framingham heart study. *Circulation* **117**, 743–753 (2008).
3. Gautam, N. Current and future applications of artificial intelligence in coronary artery disease. *Healthcare* **10**, 232 (2022).
4. Taherdoost, H. Deep learning and neural networks: Decision-making implications. *Symmetry* **15**, 1723 (2023).
5. Antoniadi, A. M. et al. Current challenges and future opportunities for XAI in machine learning-based clinical decision support systems: A systematic review. *Appl. Sci.* **11**, 5088 (2021).
6. Petmezas, G. Recent advancements and applications of deep learning in heart failure: A systematic review. *Comput. Biol. Med.* **176**, 108557 (2024).
7. Chhikara, B. S., Kumar, R., Singh, J. & Kumar, S. Heart disease prediction using machine learning and cardiovascular therapeutics development using molecular intelligence simulations: A perspective review. *Biomed Therapeutics Lett* **11**, 920–920 (2024).
8. Weng, X. et al. A joint learning method for incomplete and imbalanced data in electronic health record based on generative adversarial networks. *Comput. Biol. Med.* **168**, 107687 (2024).
9. Rehman, M. U. et al. Predicting coronary heart disease with advanced machine learning classifiers for improved cardiovascular risk assessment. *Sci. Rep.* **15**, 13361 (2025).
10. Mienye, I. D., Swart, T. G. & Obaido, G. Recurrent neural networks: A comprehensive review of architectures, variants, and applications. *Information* **15**, 517 (2024).
11. Aburass, S., Dorgham, O. & Al Shaqsi, J. A hybrid machine learning model for classifying gene mutations in cancer using lstm, bilstm, cnn, gru, and glove. *Syst Soft Comput* **6**, 200110 (2024).

12. Kingma, D. P. et al. An introduction to variational autoencoders. *Found Trends Mach Learn* **12**, 307–392 (2019).
13. Khan, S. S. et al. Novel prediction equations for absolute risk assessment of total cardiovascular disease incorporating cardiovascular-kidney-metabolic health: A scientific statement from the american heart association. *Circulation* **148**, 1982–2004 (2023).
14. Rohan, D., Reddy, G. P., Kumar, Y. P., Prakash, K. P. & Reddy, C. P. An extensive experimental analysis for heart disease prediction using artificial intelligence techniques. *Sci. Rep.* **15**, 6132 (2025).
15. Li, L. & Liu, Z.-P. Detecting prognostic biomarkers of breast cancer by regularized cox proportional hazards models. *J. Transl. Med.* **19**, 514 (2021).
16. Li, L., Zhao, W., Zhang, Q., Ching, W.-K. & Liu, Z.-P. Cnet-cox for interpretable network biomarker discovery and survival risk scoring in precise breast cancer prognosis. *npj Digital Medicine* (2026).
17. Zhang, N. Xgboost based machine learning prediction model for major adverse cardiovascular events after pci in stemi patients. Sci. Re(2026).
18. Jung, T. & Kim, J. A new support vector machine for categorical features. *Expert Syst. Appl.* **229**, 120449 (2023).
19. Kavitha, R. & Priyadharshini, M. Performance comparison of xgboost and lightgbm gradient boosting algorithms in predicting cervical cancer risk. In *2024 International Conference on Computing and Data Science (ICCDS)* 1–6 IEEE, 2024.
20. Kalabarige, L. R., Rao, R. S., Pais, A. R. & Gabralla, L. A. A boosting-based hybrid feature selection and multi-layer stacked ensemble learning model to detect phishing websites. *IEEE Access* **11**, 71180–71193 (2023).
21. Lamba, R., Gulati, T. & Jain, A. A hybrid feature selection approach for parkinson's detection based on mutual information gain and recursive feature elimination. *Arab. J. Sci. Eng.* **47**, 10263–10276 (2022).
22. Li, L., Ching, W.-K. & Liu, Z.-P. Robust biomarker screening from gene expression data by stable machine learning-recursive feature elimination methods. *Comput. Biol. Chem.* **100**, 107747 (2022).
23. Triantafyllidis, A. et al. Deep learning in mhealth for cardiovascular disease, diabetes, and cancer: Systematic review. *JMIR Mhealth Uhealth* **10**, e32344 (2022).
24. Song, Y., Ren, S., Lu, Y., Fu, X. & Wong, K. K. Deep learning-based automatic segmentation of images in cardiac radiography: A promising challenge. *Comput. Methods Progr Biomed.* **220**, 106821 (2022).
25. Kumar, G. S. & Kumaresan, P. Deep learning and transfer learning in cardiology: A review of cardiovascular disease prediction models. *IEEE Access.* **2**, 193365–193386 (2024).
26. Amirahmadi, A., Ohlsson, M. & Etminani, K. Deep learning prediction models based on ehr trajectories: A systematic review. *J. Biomed. Inform.* **144**, 104430 (2023).
27. Gonçalves, T., Rio-Torto, I., Teixeira, L. F. & Cardoso, J. S. A survey on attention mechanisms for medical applications: Are we moving toward better algorithms?. *IEEE Access* **10**, 98909–98935 (2022).
28. Tan, G. Y., Bakibillah, A., Chan, P. Y., Tan, C. P. & Nurzaman, S. Parkinson's disease tremor prediction towards real-time suppression: A self-attention deep temporal convolutional network approach. *Comput. Biol. Med.* **188**, 109814 (2025).
29. Alquraish, M., Abuhasel, A. K., Alqahtani, S. A. & Khadr, M. Spi-based hybrid hidden markov-ga, arima-ga, and arima-ga-ann models for meteorological drought forecasting. *Sustainability* **13**, 12576 (2021).
30. Silva Filho, T. Classifier calibration: A survey on how to assess and improve predicted class probabilities. *Mach. Learn.* **112**, 3211–3260 (2023).

Acknowledgements

The authors sincerely acknowledge the Faculty of Engineering and Technology (FET), Multimedia University, Melaka, Malaysia, for their valuable support and cooperation in this research.

Author contributions

A.C.D. and M.S.S. conceived of the study, developed the methodology, and registered the initial protocol. M.S.S. oversaw the data collection process. A.C.D. and N.R. were responsible for analyzing the data and grading outcomes. F.M. performed statistical analysis; A.C.D., M.S.S., and H.A.S. prepared the initial study draft. M.J.H. supervised the project. A.C.D., M.S.S., N.R., F.M., H.A.S., and M.J.H. contributed to manuscript preparation, critically revised the manuscript, and approved the final draft.

Funding

Not applicable.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to M.J.H.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026